



Kishoreganj University
3rd Year 2nd Semester B.Sc. (Engg.) Final Examination-2024
Department of Computer Science and Engineering
CSE 3203: Software Engineering (3 Credits)

Time: 3 Hours

Full Marks: 70

*Figures shown in the right margin indicate full marks.
Answer 05 out of 07 questions.*

- ✓ 1. (a) What is Software Engineering? What characteristics of software make it different from others? 4
- (b) What are the umbrella activities of a software process? 4
- ✓ (c) With an appropriate diagram, explain the spiral process model. How does this model overcome the limitations of the Waterfall model? 6
2. (a) Describe the key challenges in software engineering. 4
- ✓ (b) A team of developers has been assigned to build a Smart Health Monitoring System that uses wearable IoT devices to continuously collect and analyze health data such as heart rate, blood pressure, oxygen level, and body temperature. The system must send real-time alerts to doctors and family members in case of emergencies, and it should include a mobile app for patients to view their health trends and receive lifestyle recommendations based on AI-based predictive analytics. Doctors will also access the system through a web portal to manage patient data, provide prescriptions, and monitor health improvements over time. The system must support multiple users, secure data storage, and integration with hospital databases and government health compliance systems. The client expects the system to be deployed in multiple phases, with the core features delivered early and additional features like AI analysis and report generation added in later stages. Feedback from doctors and patients during development must be incorporated.
- i. Discuss the software development model that will be suited best for the above-mentioned scenario and justify your answer.
- ii. Differentiate the functional and non-functional requirements for given scenario.
- (c) What is agility? Explain 12 agile principles. 4
- ✓ 3. (a) Explain the four categories of software maintenance with examples. 4
- (b) Compare Maintenance Process Model-1 and Model-2. When would you prefer each model? 4
- (c) A legacy software system contains 60,000 lines of code (LOC). During one year of maintenance:
- 10,000 LOC are added
 - 4,000 LOC are deleted

- Additionally, 6,000 LOC are modified, where each modified line is considered as both one deletion and one addition.

The total development cost of the system is 6,00,000 BDT.

- i. Calculate the effective Annual Change Traffic (ACT) considering all types of changes.
 - ii. Estimate the maintenance cost using Boehm's model.
4. (a) Describe Testability? Differentiate among Testing, Quality Assurance and Quality Control. 4
- (b) A program accepts input values from 0 to 5000. Design three test cases using: 4
 - i. Equivalence Class Partitioning
 - ii. Boundary Value Analysis
- (c) Consider a ride-sharing app which has three modules: 6
 - Ride booking
 - Driver allocation
 - Payment processing

System works in unit testing but fails when a ride is booked and payment is processed together. Analyze why integration testing is critical in this scenario and evaluate whether top-down or bottom-up integration testing is more suitable.
5. (a) A game company is developing a game where players can choose different character types such as Warrior, Mage, Archer. Initially, the code uses many if-else statements to create character objects. New character types are added frequently. Analyze why the current approach is difficult to maintain. Suggest how the Factory Method pattern can solve the problem and explain the benefit. 4
- (b) A developer suggests replacing a Singleton configuration manager with a static class. Evaluate the differences between Singleton and static class. In which situations would Singleton be preferred? Give reasoning. 4
- (c) A company is building a system that creates different types of complex reports. A report may include: 6
 - Title page
 - Summary
 - Charts
 - Table data
 - Footer

Some reports require all components, while others require only a few. Propose how the Builder pattern can be applied to construct reports step-by-step. Explain why Builder is better than using multiple constructors.

6. (a) A student is building a library management system. To issue a book, the system must call multiple subsystems: 4

- Member verification
- Book availability checking
- Fine checking
- Transaction recording

Users of the system complain that the API is too complicated. Evaluate how the Façade pattern can simplify the system interface. What benefits will it bring for maintainability and usability?

- (b) A file explorer application contains: 4

- Files
- Folders

Each folder can contain more files and folders. Operations like `delete()`, `calculateSize()`, and `display()` must work for both files and folders. Analyze why treating files and folders separately increases complexity. Design a class structure using the Composite pattern to represent the file system.

- (c) A project with the following information domain characteristics: 6

Number of user inputs: 30

Number of user outputs: 55

Number of files: 24

Number of inquiries: 8

Number of external interfaces: 2

Assume that weights are complex and external complexity adjustment values are average.

- Compute the function-point value for the project.
- Find the KLOC for the project using C, Pascal and Code Generator.

✓ 7. (a) What is Cyclomatic Complexity? Consider the following code segment:

4

```
int i = 0;
while (i < n) {
    if (i % 2 == 0) {
        if (i > 10) {
            printf("A");
        } else {
            printf("B");
        }
    } else {
        printf("C");
    }
    i++;
}
```

- i) Draw the control flow graph.
- ii) Calculate the Cyclomatic Complexity $V(G)$.
- ~~iii)~~ List all independent paths.
- iv) If one more nested condition is added, predict how $V(G)$ will change and justify.

- (b) Why is risk important in Software Engineering? What is the difference between reactive and proactive risk strategies? 4
- (c) Write the differences between Verification and Validation. 3
- (d) What is meant by Complexity Adjustment Value (CAV)? What is the fundamental questionnaire of CAV? 3